RECOVERING AN OBJECT AS THE DERIVED LIMIT OF ITS TRUNCATIONS FROM A DIMENSION BOUND

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file sites-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's complete argument; the reduction and the application are both inside the counted span.

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Editorial note. The checked chapter list for same-name collisions is the set of stacks chapters staged in this repository: chow, cohomology, divisors, etale-cohomology, flat, more-algebra, more-morphisms, morphisms, spaces-more-morphisms, varieties together with this card's chapter sites-cohomology.

Editorial note. The preamble is the chapter's own retained minimal preamble (sources/stacks-sites-cohomology/retained-preamble.tex), a verbatim excerpt of the native preamble: the Xy-pic setup and the native theorem declarations with the mathematical macros.

Allowed standard prerequisites (editorial).

- Truncations and the derived limit of the system of truncations. Same chapter together with the chapter Derived Categories, the map from an object to the derived limit of its truncations. Supplies the map whose being an isomorphism is asserted.
- Vanishing in a bounded range over a covering family. Same chapter, the companion criterion and the preceding lemma of the same section, which turn a bound of the form H^p(V, H^q(E)) = 0 for p greater than some d and q negative into the hypothesis of that criterion. Supplies the reduction from the dimension bound to the criterion.

Source reference. Original remark 0H72: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5407–5407. Navigation pointer invoked at proof line 5407; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

Source reference. Original lemma 0D6N: sources/stacks-sites-cohomology/pinned-original-source.tex, source lines 5392–5392. Navigation pointer invoked at proof line 5456; the counted span contains the pointer only and the cited result is not reproduced inside it, so the pointer is registered as a non-load-bearing navigation stub in the established form for Stacks material. No bibliography citation is involved.

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Lemma 0.1. *Let $(\mathcal{C}, \mathcal{O})$ be a ringed site. Let $E \in D(\mathcal{O})$. Assume there exists an integer $d \geq 0$ and a subset $\mathcal{B} \subset \mathrm{Ob}(\mathcal{C})$ such that*

(1) *every object of $\mathcal{C}$ has a covering whose members are elements of $\mathcal{B}$,*
(2) *$H^p(V, H^q(E)) = 0$ for $p > d$, $q < 0$, and $V \in \mathcal{B}$.*

Then the map $E \to R\lim \tau_{\geq -n} E$ of Derived Categories, Remark 0H72 is an isomorphism in $D(\mathcal{O})$.

Proof. Apply Lemma 0D6N with $d_V = d$ and Cov_V equal to the set of coverings $\{V_i \to V\}$ with $V_i \in \mathcal{B}$ for all i. □